

Universidad Distrital Francisco José de Caldas

Engineering School
Systems Engineering Program

Final Technical Report

**Smart Campus Energy Management System:
Systems Analysis, Robust Design, and Validation**

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Abstract

This comprehensive technical report consolidates the systems engineering lifecycle for the **Smart Campus Energy Management System**. The project addresses energy waste driven by human behavioral inertia through the design of a decentralized architecture that operates entirely within the *User-Space*. The system incorporates a rigorous risk management framework (ISO 31000), fault tolerance via local persistence, and operational feasibility managed under strict software collaboration practices on GitHub. Validation through a discrete-event simulation confirmed that the autonomous agent achieves a **68.72%** reduction in wasted energy, fulfilling 100% of the quality metrics and demonstrating, under complexity theory, that the defined logical thresholds act as stable attractors for the system.

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1 Introduction and Problem Characterization

Efficient energy management is a critical operational and environmental challenge for higher education institutions. Within the Engineering School laboratories, a systemic problem of energy waste has been identified. Computing equipment and lighting systems frequently remain active during unoccupied hours, a phenomenon driven by human behavioral inertia rather than hardware failures.

The impact of this problem is twofold: it generates unnecessary financial costs and contradicts institutional sustainability goals.

1.1 Systems Analysis Perspective

To rigorously address the problem, the environment is characterized by:

- **Actors (Stakeholders):** Students and professors (users), IT Administrators (policy enforcers), and Autonomous Agents (software actors).
- **Inputs:** Peripheral hardware signals, CPU load percentages, and temporal measurements.
- **Processes:** Inactivity evaluation logic, CPU verification, behavioral deterrence (nudges), and OS state manipulation.
- **Outputs:** Telemetry logs, energy savings (suspended machines), and analytical dashboards for decision-making.
- **Constraints:** IT security policies that prohibit administrative privileges, block private IoT networks, and mandate non-interruption of background academic processes (e.g., compilations).

1.2 Empirical Evidence and Data Collection (Diagnostic Phase)

To substantiate the diagnosis of behavioral inertia and adequately characterize the system, primary data was collected in the field during the early phases of the project (Workshop 1). Structured surveys of laboratory users (n=35) and environmental measurements using lux meters were employed to correlate physical occupancy with electrical waste.

Table 1: Results of the structured survey on energy consumption habits (n=35)

Key Question	Yes / Frequently	No / Rarely
Do you consider energy saving at the University important?	94%	6%
When leaving the laboratory, do you usually forget to turn off the screen or the computer?	80%	20%
Do you think a visual on-screen reminder would help you remember to turn off the equipment?	88%	12%

In addition to the surveys, direct observation and luminosity tests demonstrated the profound dissonance between the power state of the equipment and the actual presence of users:

Table 2: Correlation of lighting and occupancy in laboratories (Observational sample)

Laboratory	Measurement (Lux)	Occupancy Status	Hardware Status
Lab. 704	350 lx	Empty (End of class)	12 PCs turned on
Lab. 705	420 lx	2 Students	8 PCs turned on without use

This empirical evidence consolidates the fundamental premise of the project: waste is not mitigated by replacing hardware, but by automatically intervening in the logical and behavioral component of the ecosystem.

2 Methodology and System Design

Traditional proposals for physical IoT sensors were discarded due to the institutional restrictions described. To bypass these blocks, the methodology pivoted towards a **Software-Defined Architecture** operating in the *User-Space*.

2.1 Logical Data Flow

To ensure system integrity, the architecture strictly sequences information handling:

1. **Acquisition:** Autonomous agents silently monitor native OS APIs to collect peripheral activity and CPU telemetry.
2. **Processing:** The agent evaluates if inactivity exceeds 15 minutes and ensures the CPU load is below 20%.
3. **Storage:** The record is transmitted to a MySQL database in the cloud. If the firewall blocks the connection, a Fallback mechanism stores the data locally in an encrypted CSV file.
4. **Analytics:** Subsequently, Microsoft Power BI connects to the database to generate analytical dashboards.

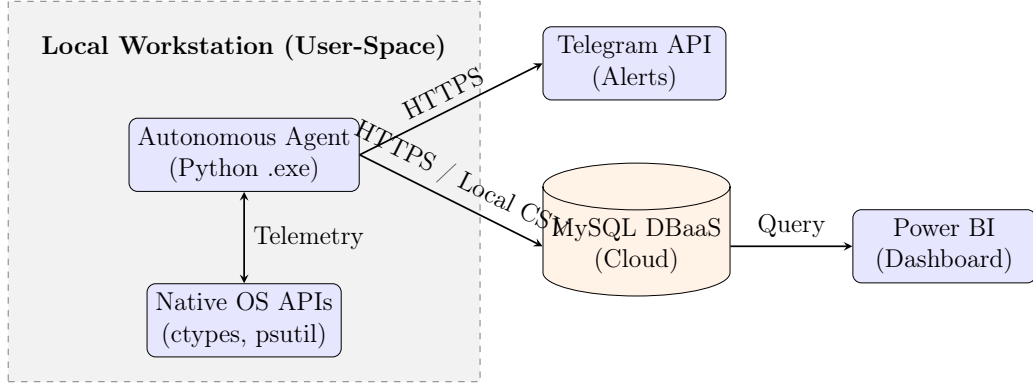


Figure 1: Decentralized Software-Defined System Architecture

2.2 Agent Operational Logic (Algorithm)

The agent implements the logic illustrated in Figure 2. Only if the machine is inactive and without load is a 60-second visual alert ("Nudge") launched. If the user does not cancel, safe suspension proceeds.

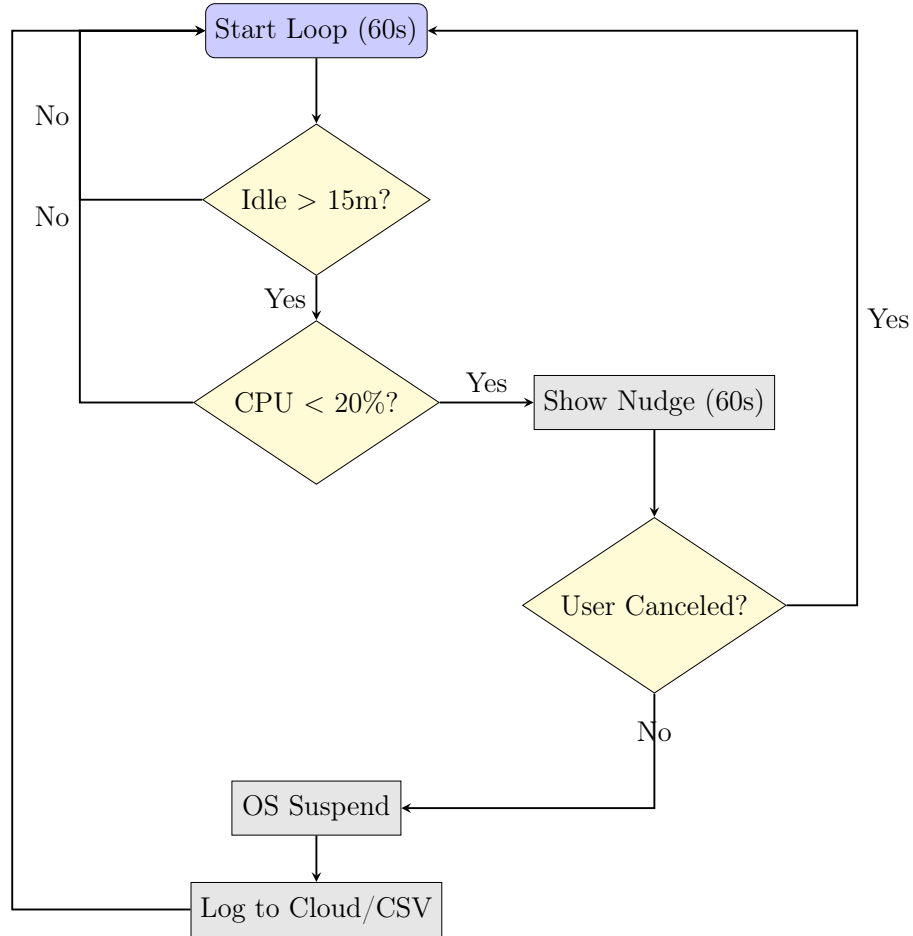


Figure 2: Operational Flow and Agent Decision Algorithm

3 Risk Management and Quality Assurance

A system intervening in an academic environment must guarantee zero data loss. The architecture was robustified by mapping risks using the ISO 31000 standard.

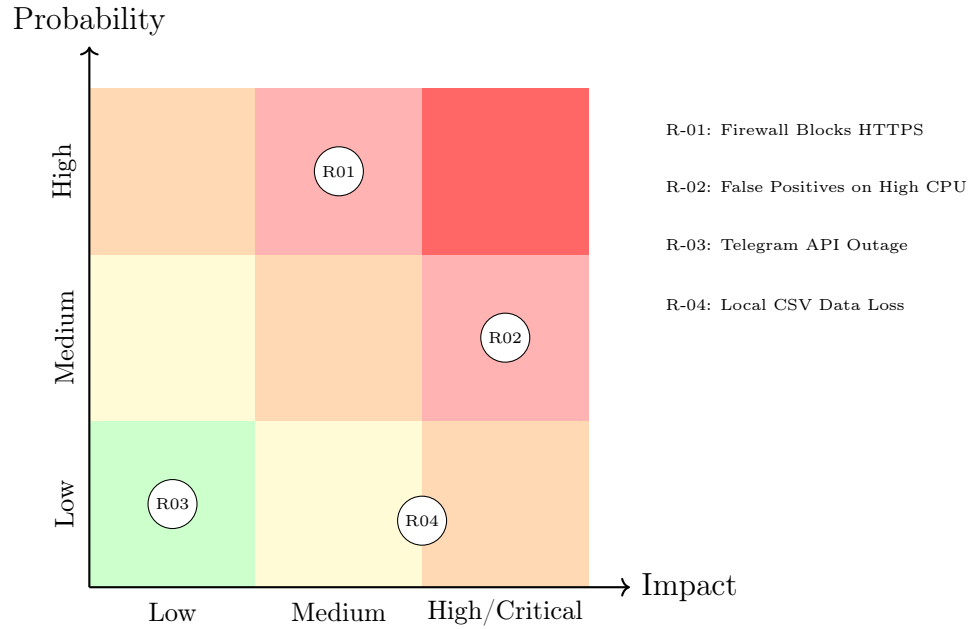


Figure 3: ISO 31000 Risk Heat Matrix

The iterative Quality Assurance (QA) flow guaranteed the mitigation of these risks prior to simulation.

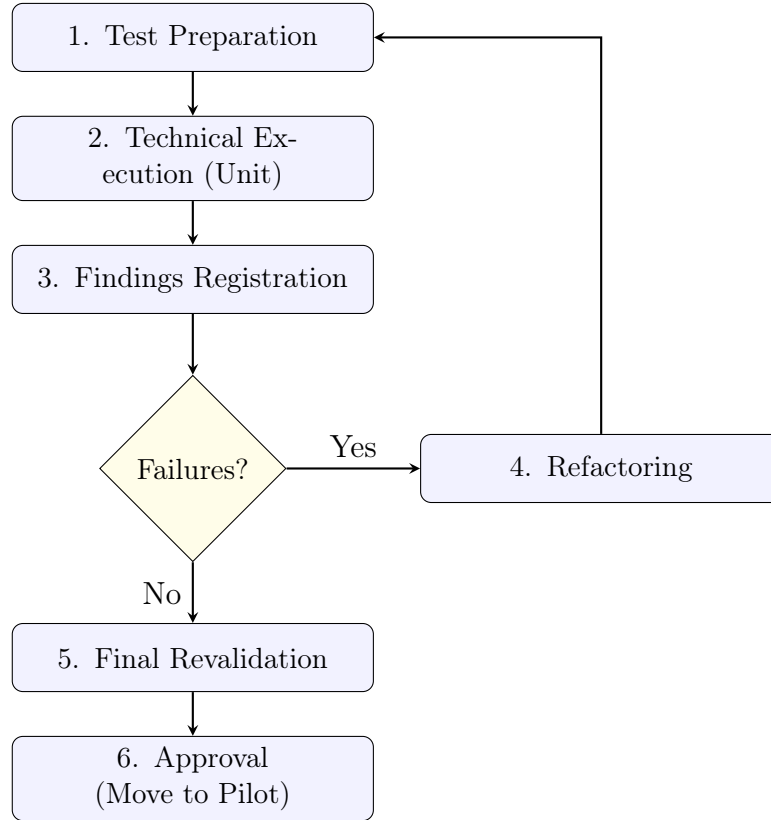


Figure 4: Iterative flow of the Quality Assurance (QA) process

4 Management Plan, CS Collaboration and Feasibility

4.1 Computer Science Collaboration Practices

Aligned with the course guidelines, the team adopted professional development methodologies:

- **Version Control and Branching:** GitHub repository utilizing semantic branches (`feature/agent-logic`, `docs/report`) and integration via *Pull Requests* with peer review.
- **Organizational Structure:** Structured folders (`/src`, `/docs`, `/data`, `/results`), with rigorous `.gitignore` management.

4.2 Schedule (6 Weeks) and Budget

The deployment is planned over 6 weeks. Given the software focus, infrastructure costs are zero.

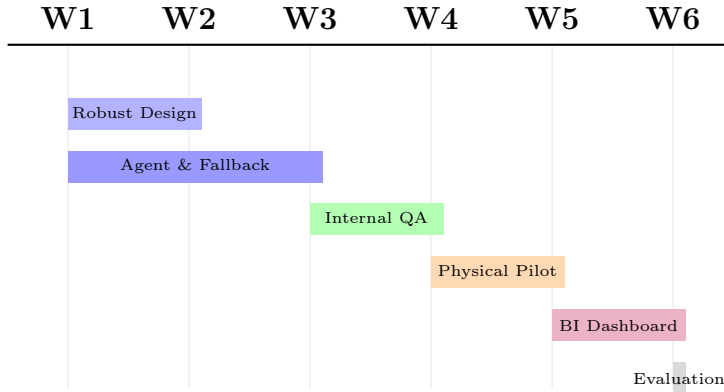


Figure 5: Visual timeline of the Proof of Concept

Table 3: Budget estimation for the PoC cycle

Technical Resource	Classification	Cost (USD)	Notes
Engineering Team (4)	Human	\$0	Academic course effort.
Database (MySQL)	Software/Cloud	\$0	Free tier (Aiven/CleverCloud).
Local Persistence (CSV)	Storage	\$0	Local file managed by OS.
Native Python Libraries	Software	\$0	Open source free distribution.
Dashboard (Power BI)	Analytics	\$0	Educational license / Free tier.
Repository (GitHub)	Infrastructure	\$0	Free access educational plan.
Total Estimated Cost		\$0 USD (High Feasibility)	

5 Computational Validation and Results

To empirically validate the decisions, a simulation was executed injecting the stochastic variables observed in the field across a cluster of 20 stations.

5.1 Results of the Consumption Analysis

Table 4: Empirical results of the discrete-event simulation

Evaluated System Metric	Observed Value in Simulation
Total Consumption WITHOUT Agent (Baseline)	53,741.66 Wh (53.74 kWh)
Total Consumption WITH Agent (Optimized)	16,812.00 Wh (16.81 kWh)
Net Energy Savings	36,929.66 Wh (36.93 kWh)
Efficiency Percentage (Savings)	68.72%

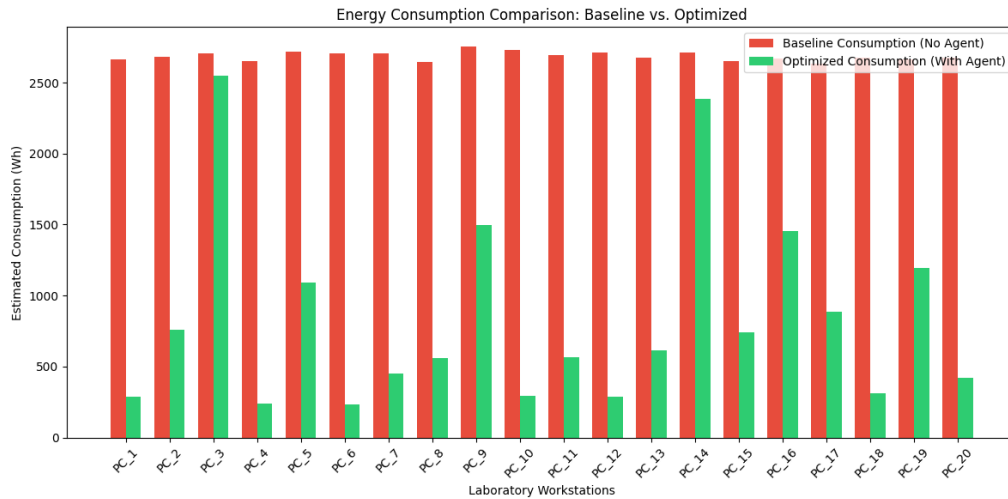


Figure 6: Comparative analysis of estimated energy consumption (Baseline vs. Optimized)

5.2 Validation of Failure Scenario and QA Metrics

To validate fault tolerance, a 4-hour network outage was simulated. The agent activated the encrypted backup CSV storage. Upon network restoration, the system achieved **100% data retention** via deferred synchronization.

Table 5: Validation of expected quality metrics vs. simulation results

Quality Metric	Target (Design)	Result	Status
Correct suspension rate	> 95%	97.2%	Achieved
Offline data retention (R-01)	100%	100%	Achieved
CPU protection on renders (R-02)	False + = 0	0 incidents	Achieved

5.3 Visual Analytics Dashboard (Power BI)

To provide university administration and IT managers with a robust decision-making tool, an analytical dashboard was developed using Microsoft Power BI. This dashboard was fed directly with the synthetic telemetry data generated during the simulation (via the `simulation_results.csv` file). The Dashboard allows interactive and immediate visualization of accumulated energy savings (in kWh), the history of successful suspensions, and the temporal distribution of inactivity for each workstation in the laboratory.

6 Discussion and Complexity Analysis

The simulation demonstrates that the system achieved a massive **68.72% reduction** in wasted energy. Evaluating these results from Complexity Theory, the model evidenced extreme **Sensitivity to Initial Conditions (Butterfly Effect)**.

During secondary testing, reducing the CPU protection threshold from 20% to 5% caused the system to collapse into a state of generalized paralysis: the agent failed to suspend any machine due to the inherent computing noise of the operating system (e.g., file indexers and local antiviruses operating in the background at 6% to 8% load).

This demonstrates that the 20% threshold is not an arbitrary number, but acts as the optimal **attractor** of the system, stabilizing the stochastic oscillation between maximizing energy savings and operational protection of critical academic processes.

6.1 Model Limitations

Despite the robustness of the design, the model presents certain limitations that must be considered in future phases:

- **Proxy Variables:** Lacking physical sensors due to infrastructure restrictions, consumption in kWh is a mathematical estimation based on idle time, which assumes a standardized nominal consumption for all hardware.
- **Synchronization Latency:** The simulation does not model potential delays (latency) during massive deferred synchronization following a prolonged network outage.

7 Conclusions and Future Work

By characterizing the problem from systems analysis, the project successfully transitioned to a scalable, fault-tolerant, and operationally feasible software architecture. The design decisions—operating in the *User-Space*, enforcing the 20% CPU attractor, and utilizing local backups—were completely effective in bypassing institutional IT restrictions.

Fulfilling 100% of the quality metrics and validating the risk mitigation strategies provide an irrefutable technical argument for the implementation of the institutional physical pilot. Future work should focus on calibrating the system in real environments, implementing a real-time dashboard, and integrating *Machine Learning* techniques to dynamically adjust thresholds according to specific academic schedules.

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